Clarifying Without Bias: Chatbots in Experimental Economics

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A standing threat to inference in experimental economics is that subjects may simply not understand the task. When choices reflect confusion about the rules or payoffs rather than underlying preferences, the resulting behavior can mimic the very objects experiments seek to measure: confusion has been shown to inflate apparent fairness, cooperation, trust, and other-regarding behavior across canonical games, and to contaminate elicited risk attitudes. Because the same data are used to build theory and to inform policy, confusion that goes unaddressed distorts both. Experimenters therefore work to clarify instructions, often relying on research assistants to answer subjects' questions during a session—but a human assistant is costly, what they tell each subject goes unrecorded and uncontrolled, and turning to a person can itself introduce bias. This motivates our question: can a large language model take over the assistant's role, reducing confusion by clarifying the rules without introducing bias by giving advice?

Answering it requires a chatbot that clarifies but never steers, and a way to measure how reliably it stays within that role. We study the question in Oprea's (2024) experiments on choice under risk, where the interpretation of behavior hinges on separating what subjects *comprehend* from how *complex* the task is to evaluate. Our chatbot may restate rules and define terms, but is forbidden to give advice, compute the best option, or reduce the task's complexity. To evaluate it without running human sessions, we have adversarial "synthetic subjects"—language models instructed to behave like difficult participants who seek advice, demand worked examples, or probe for the experimenter's intent—attack the chatbot over many fifteen-turn conversations, more than 40,000 replies in total, while an independent model scores every reply for advice-giving, factual error, and incoherence. Within this design we compare alternative chatbot architectures and base models.

Two results stand out. First, a *simpler* design is more reliable: a single, well-instructed model call adheres best, and each additional "guardrail" agent we add leaves rule-breaking unchanged or more common, because the extra steps tend to repeat the subject's forbidden request back to them. Second, the one change that clearly helps is a *stronger base model*, which cut the bias rate from about 3% to under 1% on the hardest case while also improving accuracy and speed; adding reasoning effort or a correction pass did not help. Under strict adjudication two of the three error terms nearly vanish: factual "errors" were almost entirely false positives of the first-pass judge, and complexity violations are rare on any capable model—so the only axis with appreciable signal is *bias*, and within bias the residual is not outright advice but the bot introducing expected-value or risk language. Because small per-reply risks accumulate, we report reliability over a whole conversation rather than a single turn; and because the reported bias rate counts only the conversations a cheap first-pass judge flagged for review, it is a conservative lower bound. The upshot: for a chatbot meant to clarify without biasing, the size of the base model matters more than the elaborateness of the surrounding workflow. Full methods and interactive results: `can-celebi.github.io/clarifying-without-bias`.